

Shayan Jalalipour

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Summary

Machine Learning professional and researcher specializing in computer vision, generative models, and SOTA algorithm development. Combines a strong academic publication record with hands-on experience designing and deploying end-to-end ML systems and infrastructure. Focused on driving operational success by translating cutting-edge AI research into scalable engineering solutions.

Education

Portland State University <i>PhD, Computer Science</i>	2022 - 2026
Portland State University <i>Master's Degree, Computer science</i>	2020 - 2022
Portland State University <i>Bachelor's Degree, Computer science</i>	2016 - 2020

Work experience

Portland State University <i>Machine Learning Research Assistant</i> Conducted NSF-funded interdisciplinary research by applying and evaluating new machine learning algorithms using Python, PyTorch and other advanced ML tools to publish state-of-the-art research in computer vision, generative models, and ML.	Jun 2022 - Present
PDX Intelligence <i>Founder & Principle Consultant</i> Architected and deployed production-grade machine learning pipelines, Agentic Workflows, vision models, and robust MLOps infrastructure utilizing Databricks, Docker, and WandB. Spearheaded end-to-end technical execution and client strategy, translating complex requirements into high-performance AI solutions and scalable data products. Managed full-lifecycle development from discovery to deployment while maintaining rigorous software engineering and automation standards.	Feb 2026 - Present
Handshake <i>Handshake MOVE Fellow</i> Reviewed and annotated large language model (LLM) code outputs in machine learning domains to identify, document, and mitigate behaviors compromising scientific validity, ensuring accuracy and reproducibility of model generated code. Applied advanced ML and code-review expertise to audit and annotate LLM generated source code. Identified edge case behaviors and reproducibility issues, strengthening safeguards for the scientific reliability of AI-assisted coding systems.	Oct 2025 - Present
Vacasa <i>Data Scientist</i> Collaborated with a team developing data analysis tools, creating and working with data pipelines, automating geospatial data analysis, as well as researching further applications of machine learning and AI algorithms.	Jun 2019 - Sep 2019
Portland State University <i>IT User Support</i> Part of Maseeh college of engineering IT empowering user productivity in linux / ubuntu / redhat / windows environments. Supporting students, faculty, and staff with network connectivity and management, account security, and miscellaneous IT needs.	Sep 2016 - Sep 2017

Skills

- **Languages:** Python, SQL, C++, C, Java, Javascript, Scala, Prompt Engineering
- **Libraries and Frameworks:** Pytorch, CUDA, Tensorflow, Pandas, NumPy, Kubernetes, Docker, SKLearn, HuggingFace, GNNs
- **Platforms and Services:** Git, GCP, AWS, GIS, MySQL, MongoDB, OS-terminal coding, Databricks
- **Techniques & Expertise:** Machine Learning, NLP, Statistical Knowledge, Data Visualization, Computer Vision, Deep Learning, Node.js, Transformers, Diffusion Models, Clustering, Multi-Modal Models, Reinforcement Learning, Large Language Models, LLMs

Publications

- Deep Learning-Based Spatial Detection of Drainage Structures Using Advanced Object Detection Methods..2023 **Fifth International Conference on Transdisciplinary AI (TransAI)** <https://ieeexplore.ieee.org/document/10387628>
- Noisy-Defense Variational Auto-Encoder (ND-VAE): An Adversarial Defense Framework to Eliminate Adversarial Attacks..2023 **Fifth International Conference on Transdisciplinary AI (TransAI)** <https://ieeexplore.ieee.org/document/10387596>
- OSA-Diff: An Origin Sampling Based Adversarial Attack Using Diffusion Models..2025 **19th International Conference on Semantic Computing (ICSC)** <https://ieeexplore-ieee-org.proxy.lib.pdx.edu/document/11036312>
- Understanding Adversarial Vulnerabilities and Emergent Patterns in Multimodal RL..2026 **International Conference on Semantic Computing (ICSC)** <https://ieeexplore.ieee.org/document/11486392>
- Auto-PET: PET Analysis framework for Urban Transit Video with Camera-Specific Cyclist and Pedestrian Detection Tuning..2026 **2nd International Conference on AI and Emerging Technology for Sustainable Future** (Accepted, Publication in progress)